Project: Worked examples
Title: Heat exchanger (LMTD and ε-NTU)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Single-phase heat exchanger with constant properties: in rating mode the duty and outlet temperatures for a given area; in sizing mode the area for a given duty. Counterflow, parallel flow, 1-shell-pass shell and tube, crossflow with both fluids unmixed (approximate), and one fluid at constant temperature.
Capacity rates: C_h = ṁ_h·c_p,h, C_c = ṁ_c·c_p,c, C_r = C_min/C_max Overall coefficient with fouling: U_f = 1/(1/U + R_f) Transfer units: NTU = U_f·A/C_min; effectiveness ε(NTU, C_r) for the arrangement Duty: Q = ε·C_min·(T_h,in − T_c,in); outlets from the energy balance Sizing: ε = Q/(C_min·ΔT_max), NTU from the inverse relation, A = NTU·C_min/U_f Cross-check: Q = U_f·A·F·ΔT_lm, F from R and P (1 shell pass) or back-calculated
Ref: Incropera, Fundamentals of Heat and Mass Transfer 7th ed., §11.3–11.5, Tables 11.3–11.4 — LMTD and effectiveness-NTU methods
Ref: Çengel & Ghajar, Heat and Mass Transfer 5th ed., Ch. 11 — Heat exchangers

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
arrangement	Flow arrangement	counterflow		assumption: Plate exchanger treated as pure counterflow.	any
mode	Mode	sizing (given duty)		assumption: Sizing a new exchanger.	any
m_h	Hot-side mass flow	1.2	kg/s	spec: Network connection agreement, primary flow	0.001 … 100 kg/s
cp_h	Hot-side specific heat	4190	J/(kg*K)	assumption: Water at about 70 °C.	500 … 10000 J/(kg*K)
T_h_in	Hot-side inlet temperature	80	degC	spec: Network supply temperature	-40 … 200 degC
m_c	Cold-side mass flow	1.5	kg/s	drawing: M-SCH-201 rev B	0.001 … 100 kg/s
cp_c	Cold-side specific heat	4180	J/(kg*K)	assumption: Water at about 50 °C.	500 … 10000 J/(kg*K)
T_c_in	Cold-side inlet temperature	40	degC	drawing: M-SCH-201 rev B	-40 … 200 degC
U	Overall heat transfer coefficient (clean)	3000	W/(m^2*K)	assumption: Typical clean plate exchanger, water/water.	10 … 10000 W/(m^2*K)
R_f	Total fouling resistance	0.0001	m^2*K/W	assumption: Treated closed circuits on both sides.	0 … 0.01 m^2*K/W

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
Q_req	Required duty	120	kW	spec: Building peak heat load	0.001 ... 100000 kW
dT_min	Minimum approach temperature difference	3	K	assumption: Practical approach for a plate exchanger.	0 ... 50 K

3. Assumptions

A1. Single phase on both sides (except the C_r = 0 arrangement), constant properties, no heat loss to surroundings. (Incropera 7th ed. §11.3–11.4.)
A2. The crossflow both-unmixed effectiveness is the published approximate correlation, flagged on the sheet. (Incropera 7th ed. Table 11.3.)
A3. Specific heats are user records (or B2 for air) at the mean temperature of each stream. (§A3 reference records.)
A4. arrangement (Flow arrangement) is taken as counterflow. (Plate exchanger treated as pure counterflow.)
A5. mode (Mode) is taken as sizing (given duty). (Sizing a new exchanger.)
A6. cp_h (Hot-side specific heat) is taken as 4190 J/(kg*K). (Water at about 70 °C.)
A7. cp_c (Cold-side specific heat) is taken as 4180 J/(kg*K). (Water at about 50 °C.)
A8. U (Overall heat transfer coefficient (clean)) is taken as 3000 W/(m^2*K). (Typical clean plate exchanger, water/water.)
A9. R_f (Total fouling resistance) is taken as 0.0001 m^2*K/W. (Treated closed circuits on both sides.)
A10. dT_min (Minimum approach temperature difference) is taken as 3 K. (Practical approach for a plate exchanger.)

4. Working

[12] C_h = m_h * cp_h
= 1.2 kg/s × 4190 J/(kg·K)
= 5028 W/K
Hot-side capacity rate

[13] C_c = m_c * cp_c
= 1.5 kg/s × 4180 J/(kg·K)
= 6270 W/K
Cold-side capacity rate

[14] C_min = min(C_h, C_c)
= min(5028 W/K, 6270 W/K)
= 5028 W/K
Minimum capacity rate

[15] C_r = min(C_h, C_c) / max(C_h, C_c)
= min(5028 W/K, 6270 W/K) / max(5028 W/K, 6270 W/K)
= 0.80191
Capacity ratio

[16] U_f = 1 / (1 / U + R_f)
= 1 / (1 / 3000 W/(m²·K) + 0.0001 m²·K/W)
= 2307.7 W/(m²·K)
Overall coefficient with fouling

[18] eps = Q_req / (C_min·□T_max)
= 120000 / (5028 × 40)
= 0.59666
Effectiveness required

[19] NTU = NTU(□, C_r) inverse relation — Incropera 7th ed. Table 11.4
= □ = 0.59666, C_r = 0.80191
= 1.2973
Transfer units required
[20] A_req = NTU·C_min / U_f
= 1.2973 × 5028 / 2307.7
= 2.8267 m²
Required heat transfer area
[21] UA = NTU·C_min
= 1.2973 × 5028
= 6523 W/K
Required UA
[22] Q = □·C_min·(T_h,in − T_c,in)
= 0.59666 × 5028 × 40
= 120 kW
Heat duty
[23] T_h_out = T_h,in − Q/C_h
= 80 °C − 120000 / 5028
= 56.134 degC
Hot-side outlet temperature
[24] T_c_out = T_c,in + Q/C_c
= 40 °C + 120000 / 6270
= 59.139 degC
Cold-side outlet temperature
[25] dT_c = T_c,out − T_c,in = Q/C_c
= 120000 / 6270
= 19.139 K
Cold-side temperature rise
[26] T_c_m = (T_c,in + T_c,out)/2
= (40 °C + 59.139 °C)/2
= 49.569 degC
Mean cold-side temperature
[27] dT_app = min(T_h,in − T_c,out, T_h,out − T_c,in)
= 16.134 K
= 16.134 K
Approach temperature difference
[28] dT_lm = (□T_1 − □T_2)/ln(□T_1/□T_2), counterflow ends
= □T_1 = 20.861 K, □T_2 = 16.134 K
= 18.396 K
Log-mean temperature difference
[29] F = 1 (pure counterflow, parallel flow or C_r = 0) — Incropera 7th ed. §11.3
= R = 1.247, P = 0.47847
= 1
LMTD correction factor
[30] Q_lmtd = U_f·A·F·□T_lm
= 2307.7 × 2.8267 × 1 × 18.396
= 120 kW
Duty by the LMTD method (cross-check)
[31] u_app = dT_min / dT_app
= 3 K / 16.134 K
= 0.18595
Approach utilisation (minimum / actual)

5. Results

SYMBOL	VALUE	DESCRIPTION
C_r	0.80191	Capacity ratio
NTU	1.2973	Transfer units
eps	0.59666	Effectiveness
Q	120 kW	Heat duty
T_h_out	56.134 degC	Hot outlet
T_c_out	59.139 degC	Cold outlet
dT_c	19.139 K	Cold-side rise
T_c_m	49.569 degC	Cold-side mean
cp_c	4180 J/(kg·K)	Cold-side specific heat
dT_app	16.134 K	Approach
UA	6523 W/K	UA
A_req	2.8267 m²	Required area
dT_lm	18.396 K	Log-mean temperature difference
F	1	LMTD correction factor

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Approach against minimum	u_app = 0.18595	u_app ≤ 1	0.186	

7. Applicability

WARNING — record.cp_h: cp_h: a tabulated value not taken from a reference record (Fluid specific heats); enter it as one so its source, edition and licence travel with it.
WARNING — record.cp_c: cp_c: a tabulated value not taken from a reference record (Fluid specific heats); enter it as one so its source, edition and licence travel with it.

8. Verification

Template bs.heat-exchanger@1.1.0 Content hash 9be2a4c694b4f0cd Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0